

Student Placement Prediction Using Logistic Regression

Project Objective

The objective of this project is to predict whether a student will be placed or not placed based on academic performance, skills, internships, projects, and other student-related factors.

Dataset

- Training data: 45,000 students
- Testing data: 5,000 students
- Target variable: `Placement_Status`

The dataset is already divided into training and testing sets.

Importing Libraries

```
In [44]: import pandas as pd
```

Loading Dataset

```
In [45]: train_df = pd.read_csv("train.csv")  
test_df = pd.read_csv("test.csv")
```

Understand the Data

```
In [46]: # Display the first 5 rows  
print(train_df.head())  
print(test_df.head())
```

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	\
0	1048	22	Female	B.Tech	ECE	6.29	0	3	
1	37820	20	Female	BCA	ECE	6.05	1	4	
2	49668	22	Male	MCA	ME	7.22	1	4	
3	19467	22	Male	MCA	ME	7.78	2	4	
4	23094	20	Female	B.Tech	ME	7.63	1	4	

	Coding_Skills	Communication_Skills	Aptitude_Test_Score	\
0	4	6	51	
1	6	8	59	
2	6	6	58	
3	6	6	90	
4	6	5	79	

	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	5	1	3	Not Placed
1	8	2	1	Not Placed
2	6	2	2	Not Placed
3	4	2	0	Placed
4	6	2	0	Placed

	Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	\
0	15202	23	Male	B.Tech	Civil	7.32	2	4	
1	4573	24	Female	MCA	ME	4.76	0	1	
2	34424	20	Male	BCA	ME	6.16	0	3	
3	38881	19	Male	B.Sc	IT	8.77	2	5	
4	30191	23	Male	B.Tech	ME	7.63	0	3	

	Coding_Skills	Communication_Skills	Aptitude_Test_Score	\
0	5	6	56	
1	1	4	37	
2	3	8	68	
3	8	5	83	
4	4	6	66	

	Soft_Skills_Rating	Certifications	Backlogs	Placement_Status
0	6	1	0	Placed
1	4	0	3	Not Placed
2	6	1	3	Not Placed
3	6	3	0	Placed
4	7	1	0	Not Placed

```
In [47]: # Rows x Columns
print(train_df.shape)
print(test_df.shape)
```

```
(45000, 15)
```

```
(5000, 15)
```

```
In [48]: # Datatypes
print(train_df.info())
print(test_df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 45000 entries, 0 to 44999
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Student_ID            45000 non-null  int64
1   Age                   45000 non-null  int64
2   Gender                 45000 non-null  str
3   Degree                 45000 non-null  str
4   Branch                 45000 non-null  str
5   CGPA                   45000 non-null  float64
6   Internships            45000 non-null  int64
7   Projects               45000 non-null  int64
8   Coding_Skills          45000 non-null  int64
9   Communication_Skills   45000 non-null  int64
10  Aptitude_Test_Score    45000 non-null  int64
11  Soft_Skills_Rating     45000 non-null  int64
12  Certifications         45000 non-null  int64
13  Backlogs               45000 non-null  int64
14  Placement_Status       45000 non-null  str
```

```
dtypes: float64(1), int64(10), str(4)
```

```
memory usage: 5.1 MB
```

```
None
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Student_ID            5000 non-null  int64
1   Age                   5000 non-null  int64
2   Gender                 5000 non-null  str
3   Degree                 5000 non-null  str
4   Branch                 5000 non-null  str
5   CGPA                   5000 non-null  float64
6   Internships            5000 non-null  int64
7   Projects               5000 non-null  int64
8   Coding_Skills          5000 non-null  int64
9   Communication_Skills   5000 non-null  int64
10  Aptitude_Test_Score    5000 non-null  int64
11  Soft_Skills_Rating     5000 non-null  int64
12  Certifications         5000 non-null  int64
13  Backlogs               5000 non-null  int64
14  Placement_Status       5000 non-null  str
```

```
dtypes: float64(1), int64(10), str(4)
```

```
memory usage: 586.1 KB
```

```
None
```

```
In [49]: # Checking Number of Students Placed and Not Placed
print(train_df['Placement_Status'].value_counts())
```

```
Placement_Status
Not Placed    28688
Placed        16312
Name: count, dtype: int64
```

```
In [50]: # Checking percentage of students placed
print(train_df['Placement_Status'].value_counts(normalize=True) * 100)
```

```

Placement_Status
Not Placed    63.751111
Placed        36.248889
Name: proportion, dtype: float64

```

Data Quality Check

```

In [51]: # Checking Null Values
print(train_df.isnull().sum())
print(test_df.isnull().sum())

```

```

Student_ID      0
Age             0
Gender          0
Degree          0
Branch          0
CGPA            0
Internships     0
Projects        0
Coding_Skills   0
Communication_Skills 0
Aptitude_Test_Score 0
Soft_Skills_Rating 0
Certifications  0
Backlogs        0
Placement_Status 0
dtype: int64
Student_ID      0
Age             0
Gender          0
Degree          0
Branch          0
CGPA            0
Internships     0
Projects        0
Coding_Skills   0
Communication_Skills 0
Aptitude_Test_Score 0
Soft_Skills_Rating 0
Certifications  0
Backlogs        0
Placement_Status 0
dtype: int64

```

```

In [52]: # Checking Duplicates
print("Train duplicates:", train_df.duplicated().sum())
print("Test duplicates:", test_df.duplicated().sum())

```

```

Train duplicates: 0
Test duplicates: 0

```

Data Preprocessing

```

In [53]: # Define Numerical & Categorical Columns
numerical_cols = ['Age', 'CGPA', 'Internships', 'Projects', 'Coding_Skills', 'Co

categorical_cols = ['Gender', 'Degree', 'Branch']

```

```
In [54]: # Column Transformer
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), numerical_cols),
        ('cat', OneHotEncoder(drop='first', handle_unknown='ignore'), categorical_cols)
    ]
)
```

Separating Features and Target

```
In [55]: # Features
X = train_df.drop(['Placement_Status', 'Student_ID'], axis=1)

# Target
y = train_df['Placement_Status']
```

```
In [56]: # Features
X_test = test_df.drop(['Placement_Status', 'Student_ID'], axis=1)

# Target
y_test = test_df['Placement_Status']
```

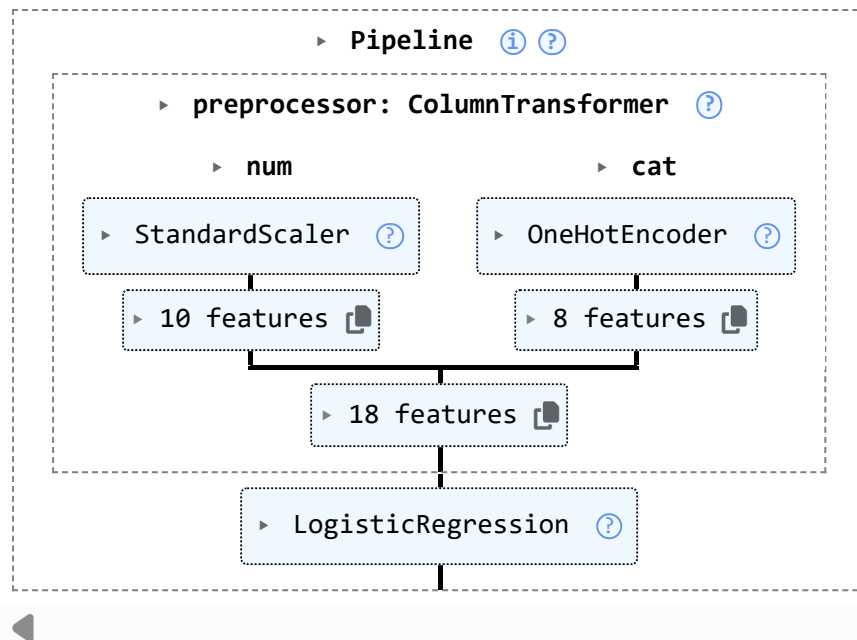
Logistic Regression Model

```
In [57]: # Importing Pipeline
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
```

```
In [58]: model = Pipeline([
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression())
])
```

```
In [59]: # Training the Model
model.fit(X, y)
```

Out[59]:

In [60]: `y_pred = model.predict(X_test)`

Model Evaluation

In [61]: `from sklearn.metrics import accuracy_score, confusion_matrix, classification_report`

```

In [62]: accuracy = accuracy_score(y_test, y_pred) # Accuracy
cm = confusion_matrix(y_test, y_pred) # Confusion Matrices

print("Accuracy:", accuracy)
print(cm)
print(classification_report(y_test, y_pred)) # Classification Report

```

Accuracy: 0.8722

[[2869 319]

[320 1492]]

	precision	recall	f1-score	support
Not Placed	0.90	0.90	0.90	3188
Placed	0.82	0.82	0.82	1812
accuracy			0.87	5000
macro avg	0.86	0.86	0.86	5000
weighted avg	0.87	0.87	0.87	5000

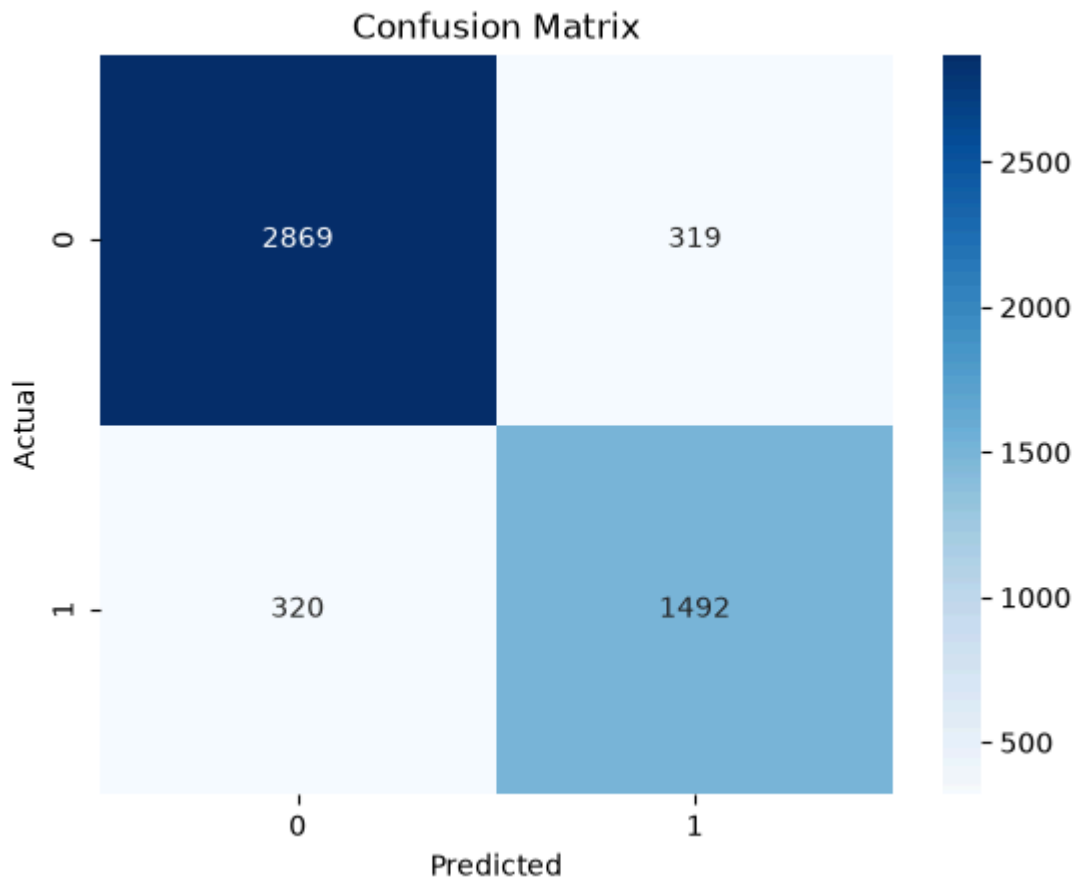
```

In [63]: # Visualization
import seaborn as sns
import matplotlib.pyplot as plt

# using Heatmap
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

```



Predict New Student

```
In [64]: # new_student info
new_student = pd.DataFrame({
    'Age': [22],
    'Gender': ['Male'],
    'Degree': ['B.Tech'],
    'Branch': ['Computer Science'],
    'CGPA': [8.2],
    'Internships': [2],
    'Projects': [3],
    'Coding_Skills': [8],
    'Communication_Skills': [8],
    'Aptitude_Test_Score': [82],
    'Soft_Skills_Rating': [8],
    'Certifications': [3],
    'Backlogs': [0]
})
```

```
In [65]: # Predicting new_students' placement
prediction = model.predict(new_student)

print("Placement Prediction:", prediction[0])
```

Placement Prediction: Placed

C:\Users\Sandeep\AppData\Roaming\Python\Python311\site-packages\sklearn\preprocessing_encoders.py:262: UserWarning: Found unknown categories in columns [2] during transform. These unknown categories will be encoded as all zeros
warnings.warn(msg, UserWarning)

```
In [66]: # Probability of getting new_student placed
probability = model.predict_proba(new_student)

print("Not Placed:", probability[0][0])
print("Placed:", probability[0][1])
```

Not Placed: 0.005647192366877385

Placed: 0.9943528076331226

```
C:\Users\Sandeep\AppData\Roaming\Python\Python311\site-packages\sklearn\preprocessing\_encoders.py:262: UserWarning: Found unknown categories in columns [2] during transform. These unknown categories will be encoded as all zeros
warnings.warn(msg, UserWarning)
```

Conclusion

A Logistic Regression model was developed to predict student placement status using academic performance, skills, internships, projects, and other student-related features.

The data was preprocessed using `ColumnTransformer`, with `StandardScaler` applied to numerical features and `OneHotEncoder` applied to categorical features. The preprocessing and Logistic Regression model were combined using a Pipeline.

The model was evaluated using accuracy, confusion matrix, precision, recall, and F1-score. The trained model can also predict the placement status and placement probability for a new student.